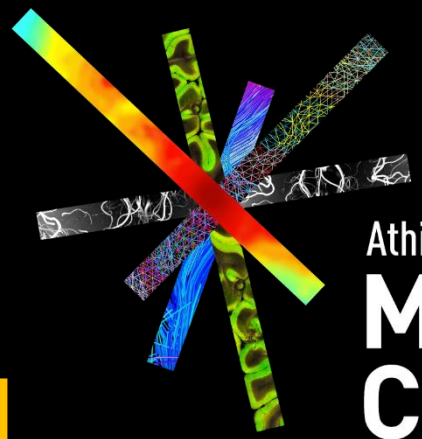
SeqEyes: sequence and trajectory viewer tool



Athinoula A.
Martinos
Center
For Biomedical Imaging

Xingwang Yong

Athinoula A. Martinos Center for Biomedical Imaging, Massachusetts General Hospital, Charlestown, Massachusetts, USA

Why do we need a .seq viewer?

- seq.plot() is useful, works perfectly fine for small sequence
- .seq files can be large and complex, multi echo/shot/average/repetition…
- Challenges in current workflow
  - **Slow for large files**: Slow to interact
  - No built-in **time-synchronized** k-space trajectory visualization
  - **Manual, command-based time window adjustment**: Viewing a specific time window requires typing seq.plot('TimeRange',…), every adjustment means re-typing
- These limitations prevent fast sequence debugging

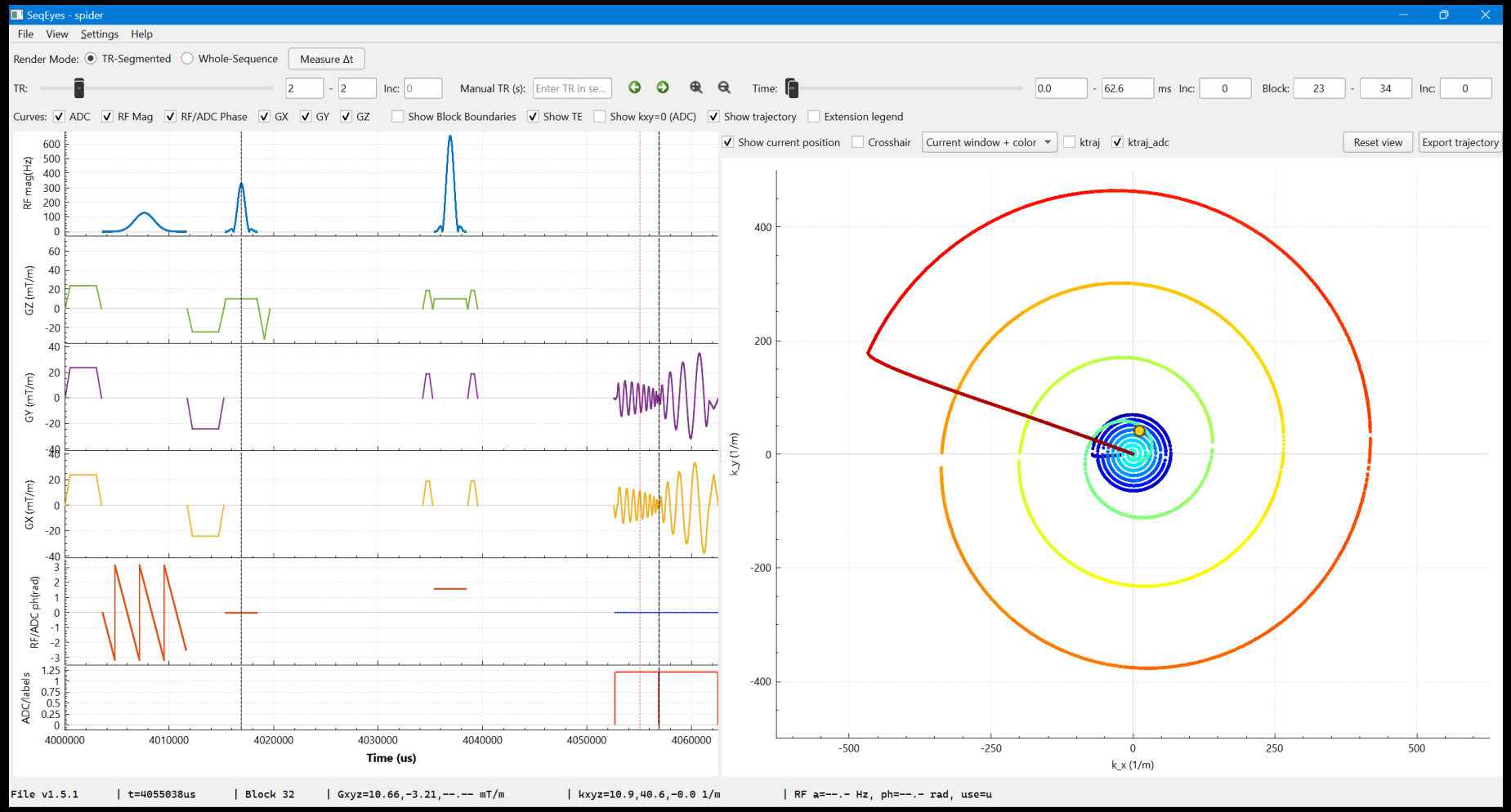
How SeqEyes addresses these challenges

Challenges	SeqEyes solution
Slow rendering and interaction	▪ C++ with Qt ▪ Undersample complicated curves
No built-in synced trajectory	Trajectory view, synced with sequence diagram
Repeated seq.plot(‘TimeRange’, …)	▪ Mouse wheel to zoom ▪ Drag to pan ▪ Navigation based on TR index/block index/time

➢ Key Characteristics of SeqEyes

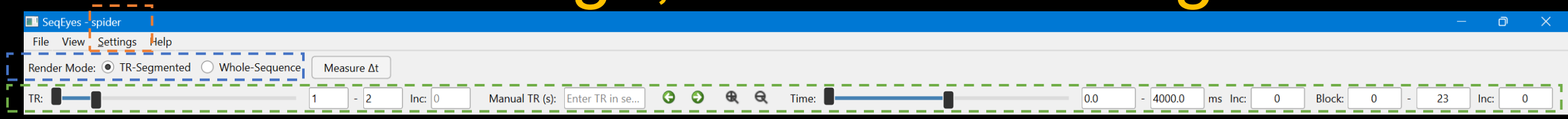
- Standalone desktop application
- Cross-platform (windows, linux)
- Performance-oriented, interactive UI

SeqEyes UI overview



Example spin-echo sequence: fat saturation, RF 90, RF 180, spiral-in, spiral-out

UI walkthrough, controls & navigation



➢ Menu bar: settings

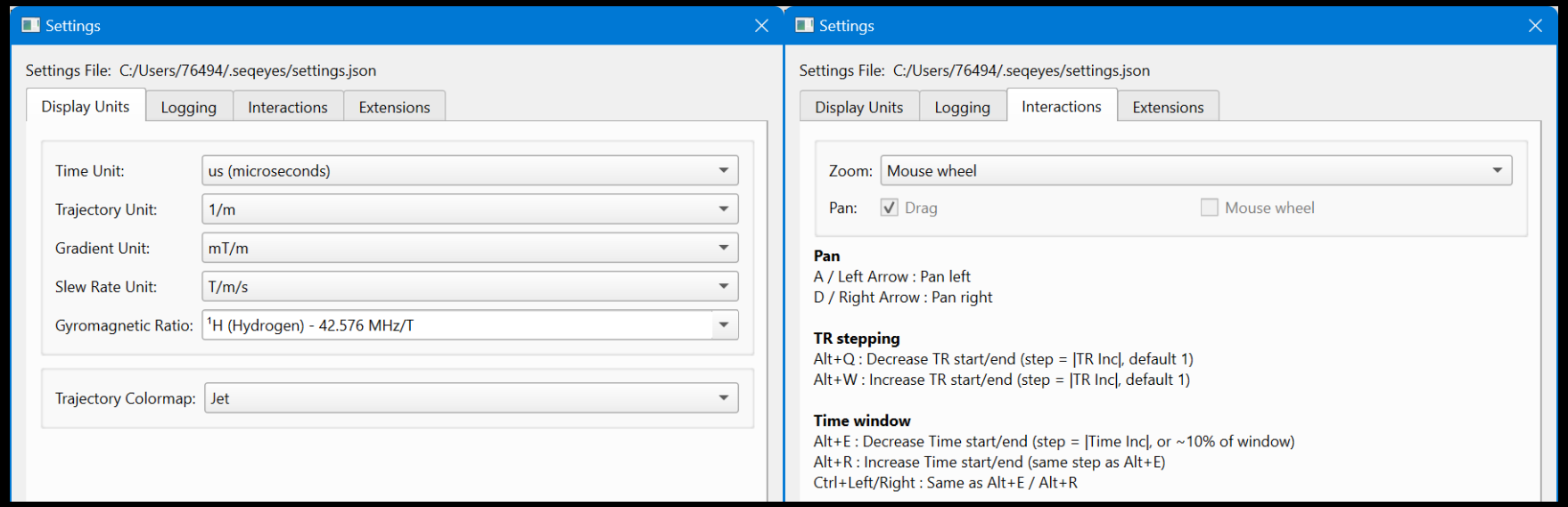
- Gradient/trajectory/time unit
- Preferred way of zoom/pan
- Extension display control

➢ Render mode

- Whole-sequence, =seq.plot()
- TR-segmented

➢ Navigation

- TR-based, =seq.plot('TimeRange',TR*[1 2])
- Time-based, =seq.plot('TimeRange',[1 2])
- Block-index-based, =seq.plot('BlockRange',[1 2])

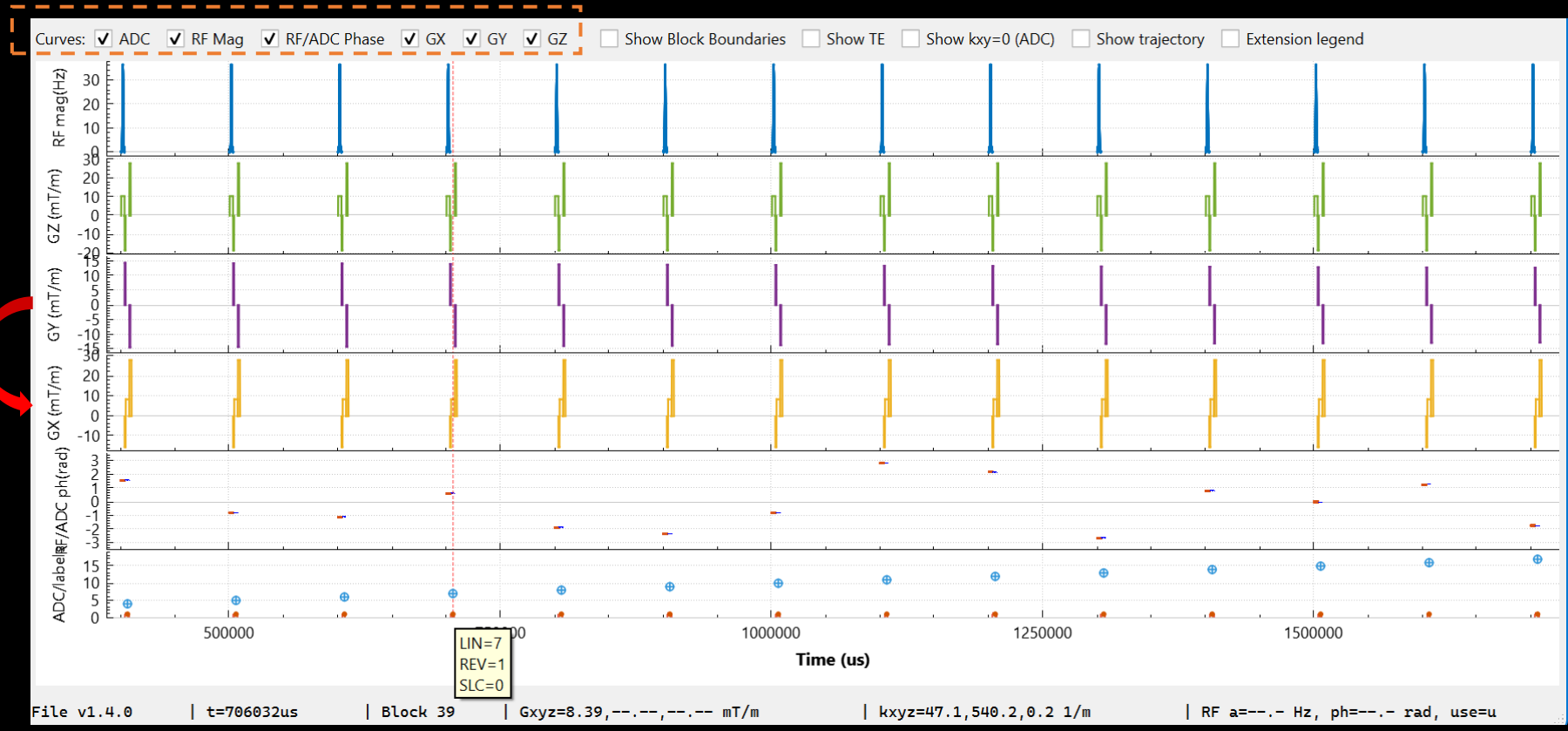


UI walkthrough, sequence diagram

➢ Channel control

- Channels can be enabled/disabled
- Channels can be reordered
- Extensions plotted on ADC channel

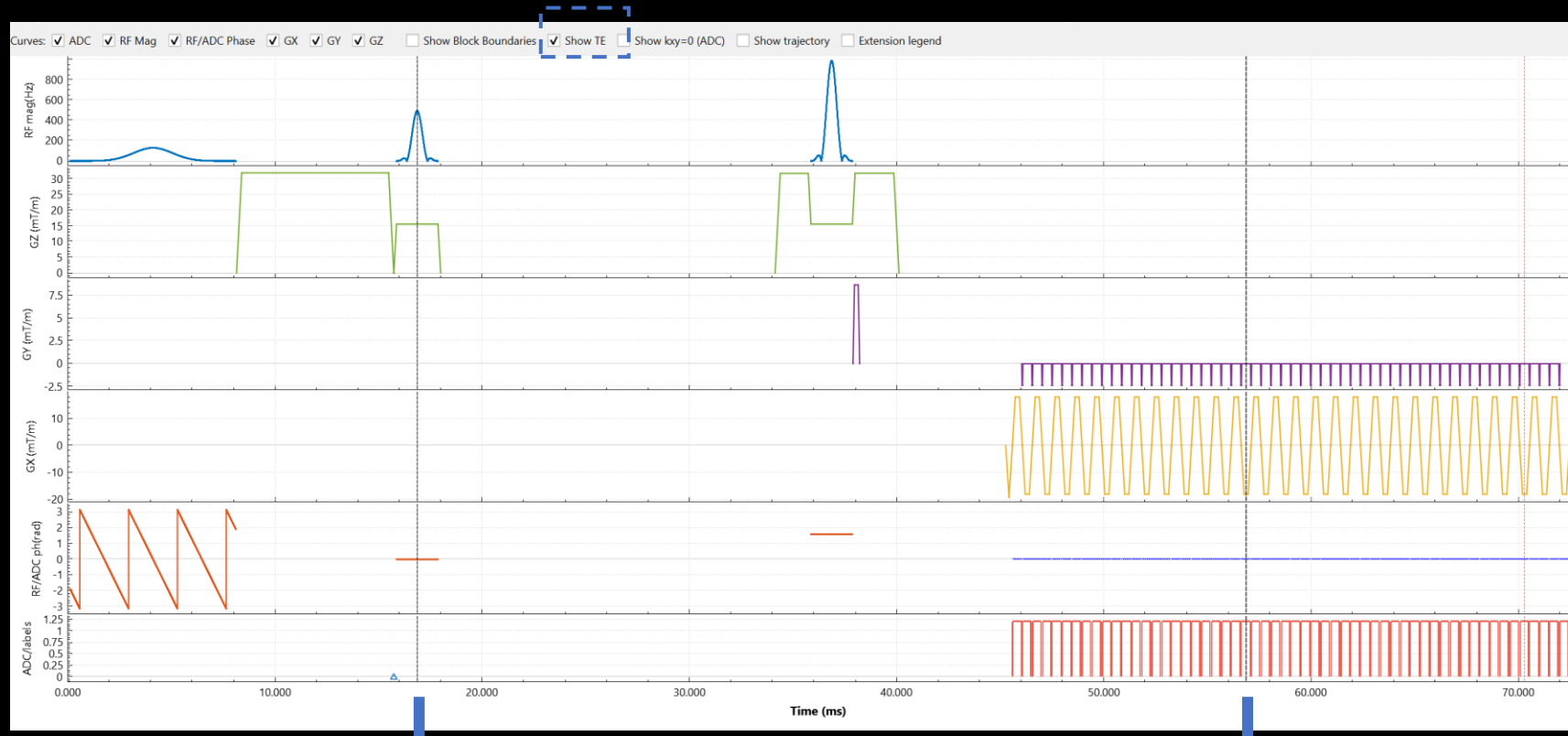
drag&drop channel title to reorder



UI walkthrough, sequence diagram

➢ Optional overlay

- TE overlay: 2 lines at excitation RF center and TE



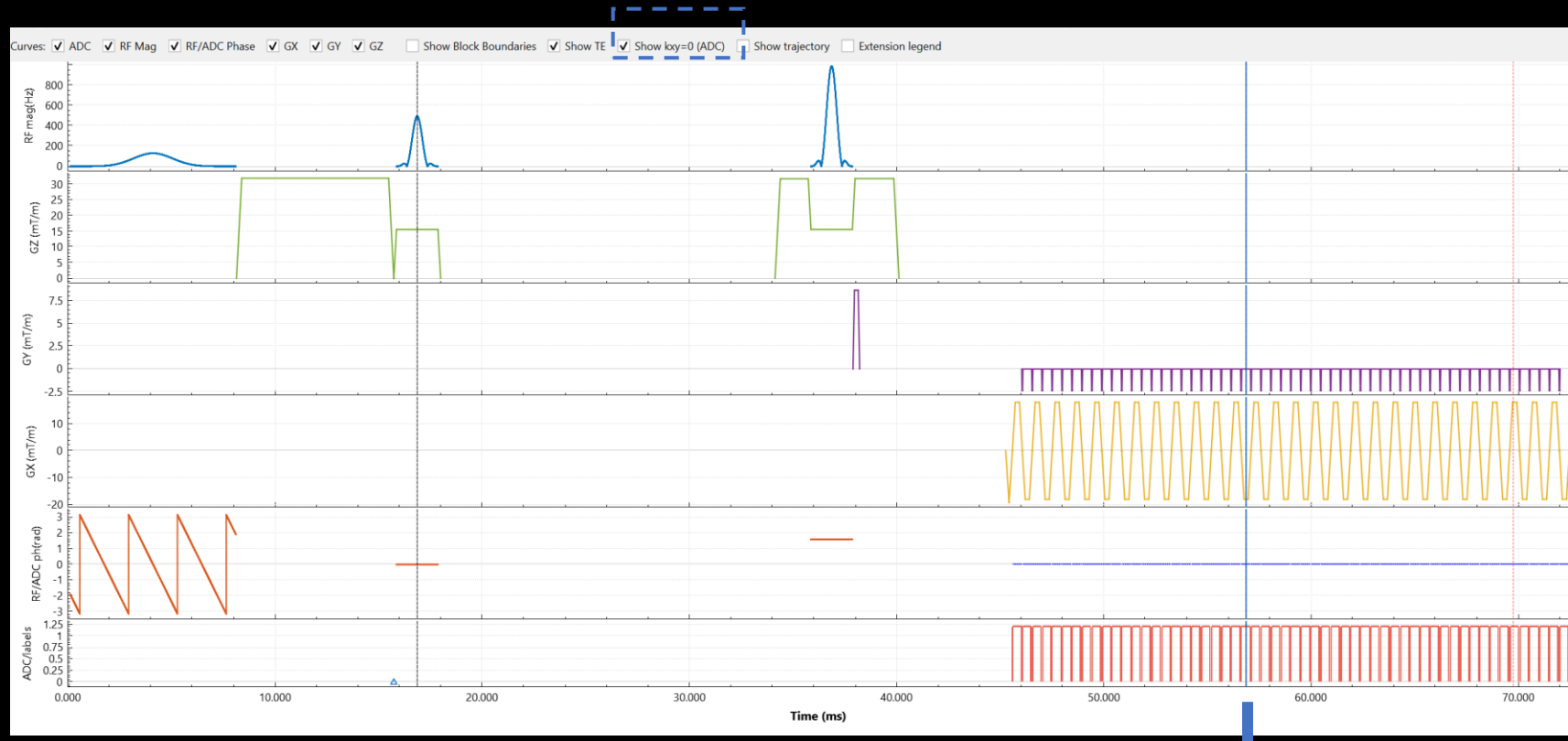
RF center

t=TE

UI walkthrough, sequence diagram

➢ Optional overlay

- TE overlay: 2 lines at excitation RF center and TE
- kxy=0 overlay: line at kx=ky=0

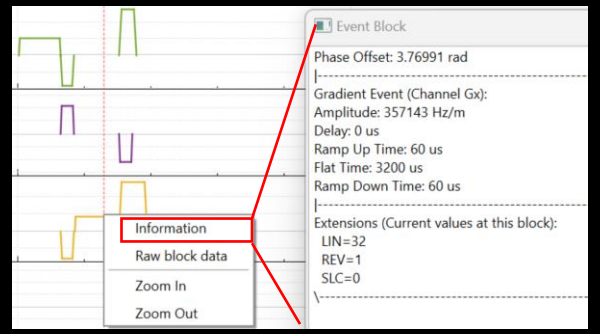


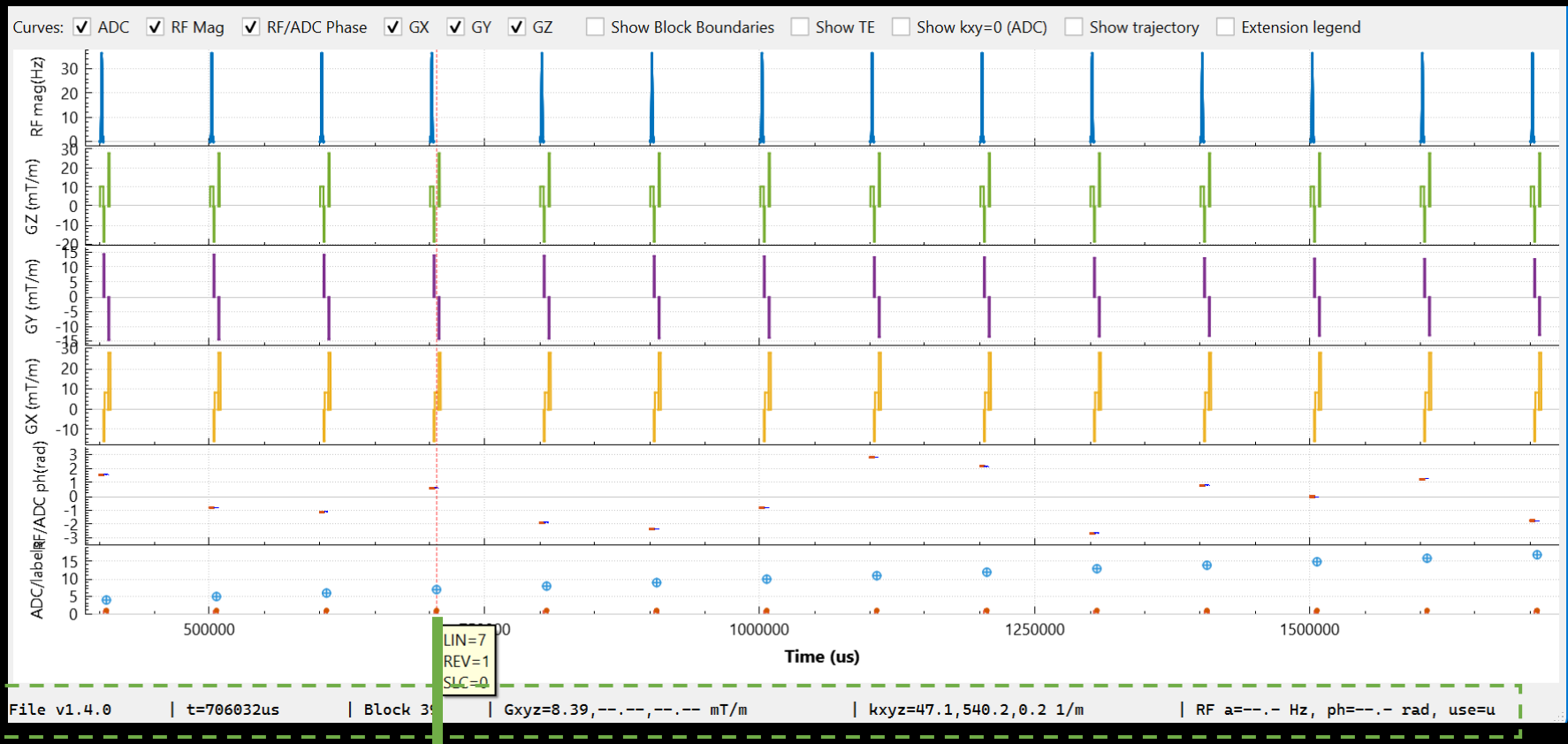
Overlapped lines: t=TE, kx=ky=0

UI walkthrough, sequence diagram

➢ Information window

- Right click to show decoded info





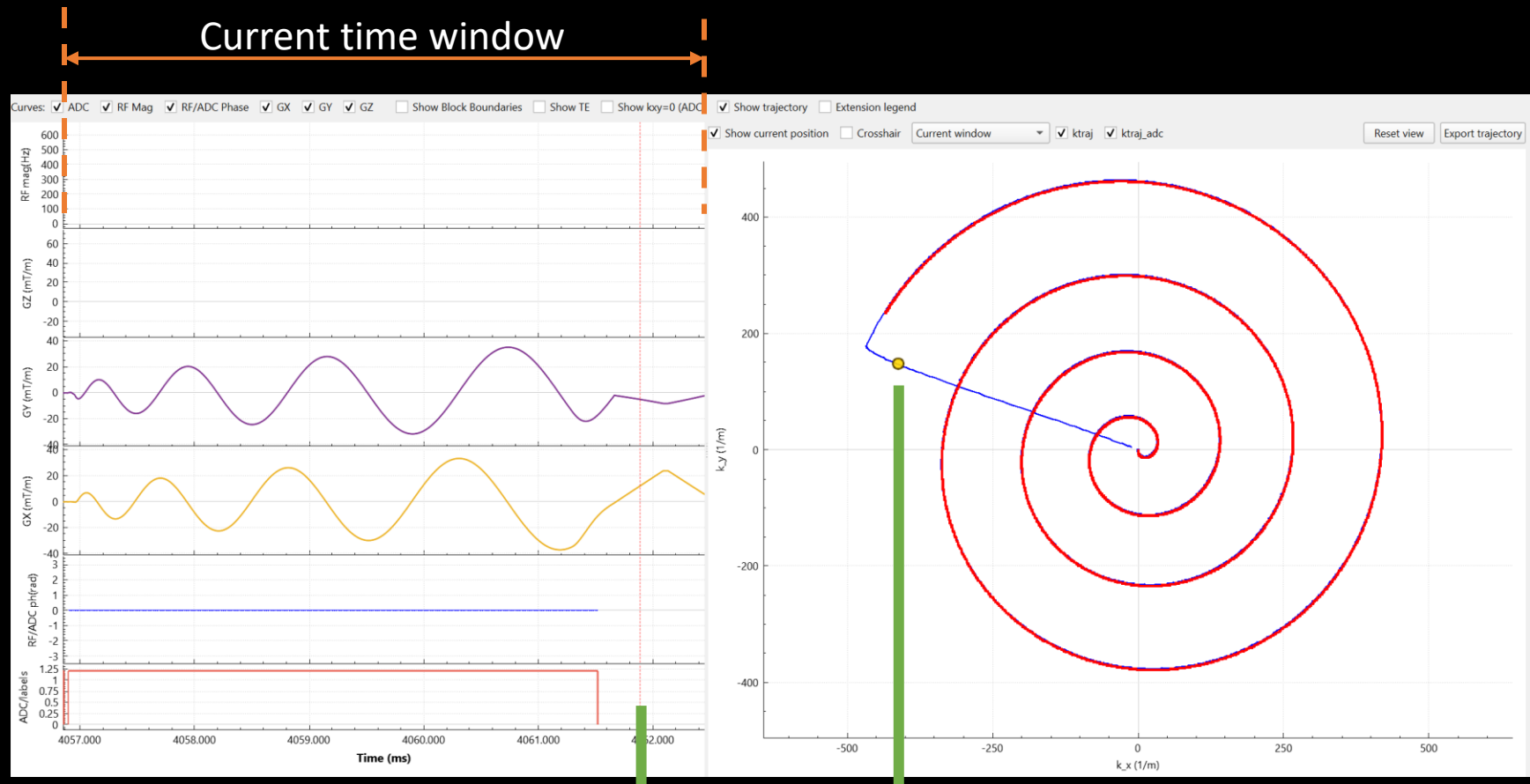
Red dashed line: current mouse position

➢ Status bar

- .seq file version
- current time
- current block
- current gradient
- current kxyz
- current RF

UI walkthrough, k-space trajectory

- Synced with sequence diagram
  - By default, show trajectory of the current time window in the sequence diagram
- Cursor sync
  - Move mouse in sequence diagram, yellow circle moves in trajectory

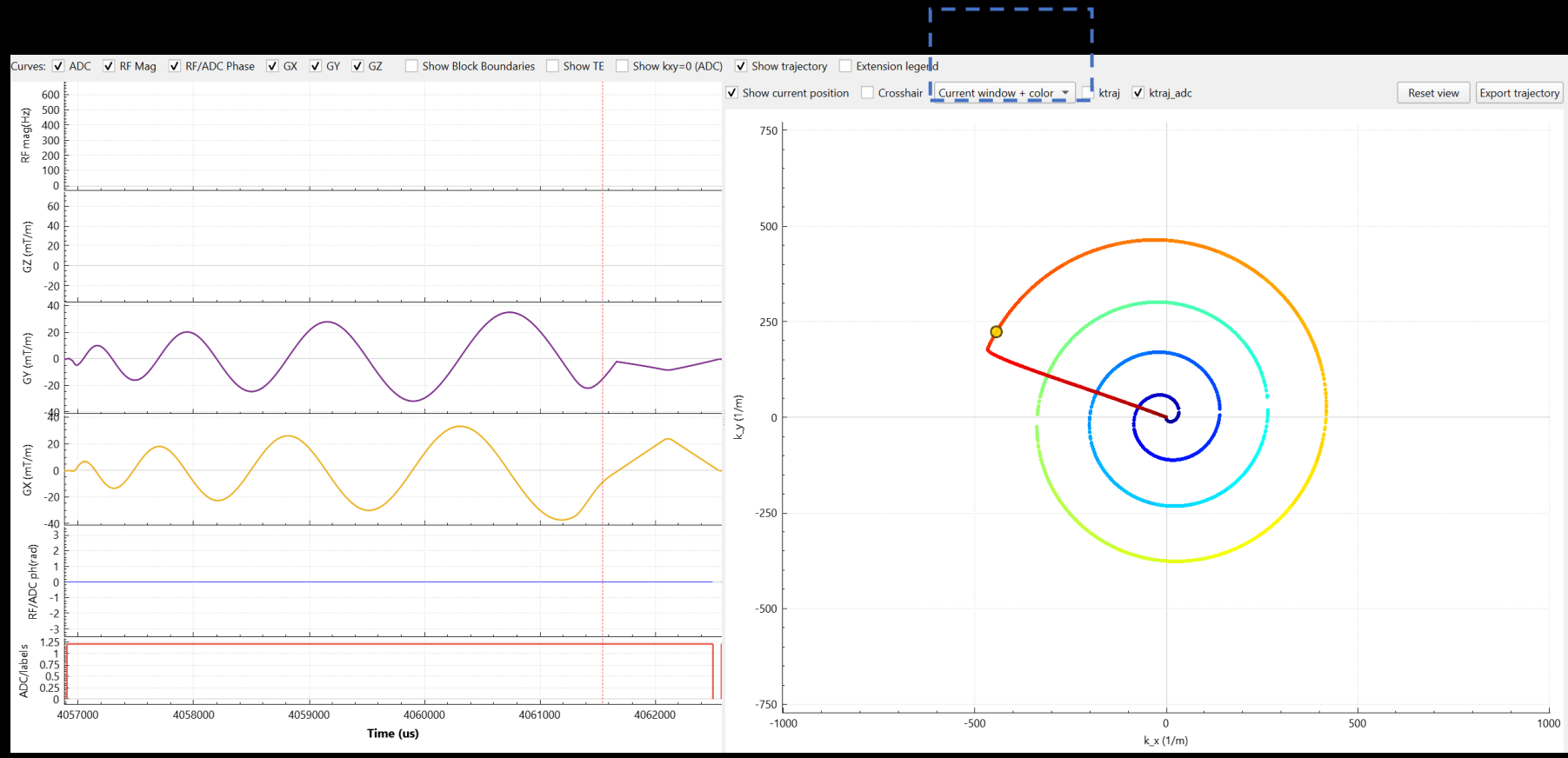


Red dashed line: current mouse position

Yellow circle: corresponding k-space position

UI walkthrough, k-space trajectory

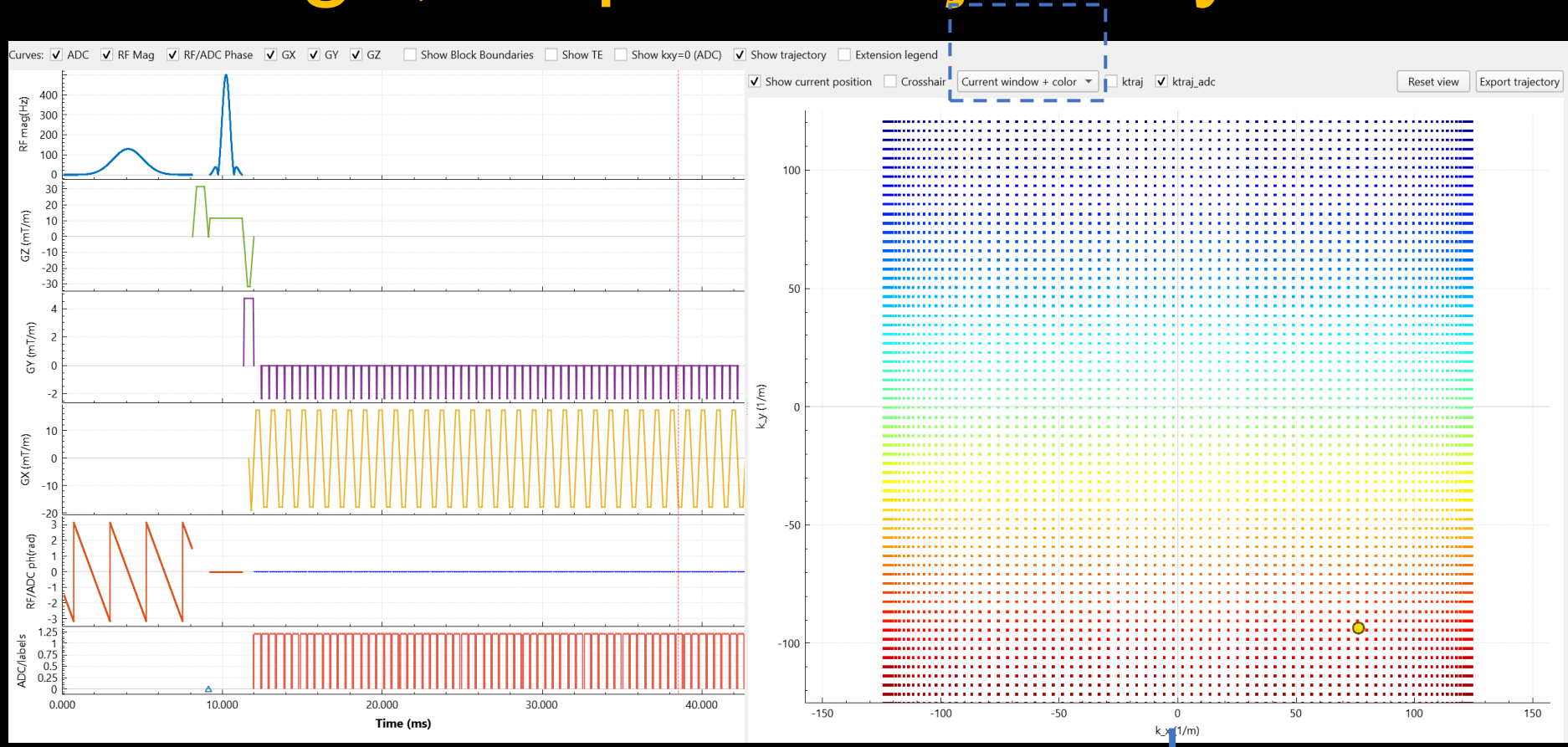
- Color encodes time (jet colormap)
  - Blue=early
  - Red=late



Example spiral-out readout

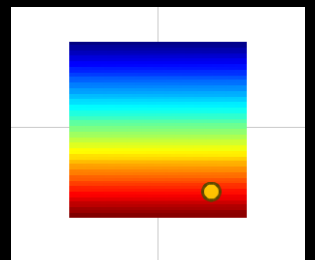
UI walkthrough, k-space trajectory

- Color encodes time (jet colormap)
  - Blue=early
  - Red=late



demoSeq/writeEpiRS.m

zoom out



Matlab wrapper & CLI

- Matlab

```
>>seqeyes('file.seq')
>>seqeyes(seq)
>>seqeyes `seq`
>>seqeyes --layout 211 file.seq
```

- CLI

```
$seqeyes file.seq
$seqeyes --layout 211 file.seq
$seqeyes --TR-range 1~5 file.seq
$seqeyes --no-RFphase --no-Gx file.seq
```

```
>> seqeyes --help
Command: "C:\Users\76494\Downloads\New folder (22)\seqeye\out\build\x64-Rele
Usage: C:\Users\76494\Downloads\New folder (22)\seqeye\out\build\x64-Release
SeqEyes - Pulseq Sequence Viewer

Options:
  -h, --help                    Show this help text and exit
  -v, --version                 Show version information and exit
  --maximized                   Start maximized
  --name <title>                Set window title to the specified name
  --no-ADC                      Hide ADC/labels axis
  --no-RFmag                    Hide RF magnitude axis
  --no-RFphase                  Hide RF/ADC phase axis
  --no-Gx                       Hide Gx axis
  --no-Gy                       Hide Gy axis
  --no-Gz                       Hide Gz axis
  --TR-segmented                TR-Segmented Mode
  --Whole-sequence              Whole-Sequence Mode
  --TR-range <start~end>        TR range as start~end (1-based)
  --time-range <start~end>      Time range as start~end in ms
  --layout <abc>                Subplot layout as abc (e.g., 211, Matlab
                                subplot style)
  --headless                    Do not show GUI (for testing/CLI)
  --exit-after-load             Exit after loading file (no event loop).
                                Implies --headless.
  --automation <scenario.json>  Run automation scenario JSON (implies
                                --headless)

Arguments:
  file                          Pulseq sequence file (.seq) to open
```

Athinoula A.
Martinos
Center
For Biomedical Imaging

Thank you!

Resources

- https://github.com/xingwangyong/seqeyes
- https://github.com/xpjiang/PulseqViewer

Acknowledgement

- Berkin Bilgic
- Yuting Chen
- Yiyun Dong
- Paul Dubovan
- Yohan Jun
- R01-EB032378